Chemistry Formula Sheet (College & Pre-Med)

General and organic chemistry for college, pre-med and MCAT review, plus the clinical chemistry med school builds on

MindShelf

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Units & Concentration

$$M = \frac{n}{V_{\text{soln}}} \quad m = \frac{n_{\text{solute}}}{m_{\text{solvent (kg)}}} \quad n = \frac{m}{\mathcal{M}}$$

$C_1V_1 = C_2V_2$ eq $= n \times |z|$ osmolarity $= i \times M$

$\mathcal{M}$ molar mass; i van 't Hoff factor; z ion charge. *E.g.* 500 mL of 0.1 M from 2 M stock: $V_1 = 0.1 \times 500/2 = 25$ mL.

Atoms & Periodic Trends

$$A = Z + N \quad E = h\nu = \frac{hc}{\lambda} \quad c = \lambda\nu$$

$n \geq 1$; $\ell = 0 \ldots n-1$; $m_\ell = -\ell \ldots +\ell$; $m_s = \pm\frac{1}{2}$.

Across a period →: radius ↓, ionization energy ↑, electronegativity ↑; down a group ↓: reversed.

Bonding & Geometry

Formal charge

$$\text{FC} = V - N - \frac{B}{2}$$

V valence, N nonbonding, B bonding e⁻. Best: charges near zero, minus on the more electronegative atom.

Dom.	LP	Shape (e.g.)	Angle	Hybrid
2	0	linear (CO_2)	180°	sp
3	0	trig. planar (BF_3)	120°	sp^2
3	1	bent (SO_2)	<120°	sp^2
4	0	tetrahedral (CH_4)	109.5°	sp^3
4	1	trig. pyramidal (NH_3)	≈107°	sp^3
4	2	bent (H_2O)	≈104.5°	sp^3
5	0	trig. bipyramid (PCl_5)	90, 120°	sp^3d
6	0	octahedral (SF_6)	90°	sp^3d^2

IMFs: ion–dipole > H-bond (H on N, O, F) > dipole–dipole > London. Stronger IMFs: higher bp, lower vapor pressure.

Gases

$$PV = nRT \quad \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$P_i = X_iP_{\text{tot}} \quad \frac{r_1}{r_2} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}}$$

T in kelvin. *E.g.* 2 atm, 5 L, 300 K, $R = 0.08206$: $n = PV/RT \approx 0.406$ mol.

Thermochemistry

$$q = mc\Delta T \quad \Delta G = \Delta H - T\Delta S$$

$$\Delta H_{\text{rxn}} = \sum \Delta H_f^{\text{prod}} - \sum \Delta H_f^{\text{react}}$$

$$\Delta G^\circ = -RT \ln K = -nFE^\circ$$

E.g. $\Delta H = -92.2$ kJ/mol, $\Delta S = -0.199$ kJ/(mol·K), 298 K: $\Delta G \approx -32.9$ kJ/mol.

Spontaneous ($\Delta G < 0$): $\Delta H < 0, \Delta S > 0$ always; $\Delta H > 0, \Delta S < 0$ never; both negative at low T; both positive at high T.

Kinetics & Enzymes

$$\text{rate} = k[A]^m[B]^n \quad k = Ae^{-E_a/RT}$$

First order: $\ln[A] = \ln[A]_0 - kt$; $t_{1/2} = 0.693/k$.

$$v = \frac{V_{\max}[S]}{K_m + [S]} \quad \frac{1}{v} = \frac{K_m}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}$$

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Orders come from experiment, not coefficients. *E.g.* K_m = 2 mM, [S] = 6 mM, V_{max} = 100: v = 75 µmol/min.

Inhibitor	Binds	K_m	V_{max}
Competitive	E (active site)	↑	=
Noncompetitive	E and ES equally	=	↓
Uncompetitive	ES only	↓	↓
Mixed	E or ES, unequally	↑ / ↓	↓

Equilibrium & Solubility

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b} \qquad K_{sp}\left(M_xX_y\right) = [M]^x[X]^y$$

$Q < K$: shifts right; $Q > K$: shifts left; solid forms if $Q > K_{sp}$. Common ion: lower solubility. Solids and pure liquids stay out of K.

Acids, Bases & Buffers

$$\text{pH} = -\log[H_3O^+] \qquad K_w = [H_3O^+][OH^-]$$

$$pK_a + pK_b = pK_w \qquad \text{pH} = pK_a + \log\frac{[A^-]}{[HA]}$$

Best buffer within $pK_a \pm 1$. Weak acid titrated by strong base: half-equivalence pH = pK_a; equivalence pH > 7.

Disorder	pH	Primary	Comp.
Metabolic acidosis	↓	↓ HCO_3^-	↓ $PaCO_2$
Metabolic alkalosis	↑	↑ HCO_3^-	↑ $PaCO_2$
Respiratory acidosis	↓	↑ $PaCO_2$	↑ HCO_3^-
Respiratory alkalosis	↑	↓ $PaCO_2$	↓ HCO_3^-

CLINICAL NOTE Bicarbonate buffer

$$\text{pH} = 6.1 + \log\frac{[HCO_3^-]}{0.03 \times P_aCO_2}$$

HCO_3^- 24 mmol/L, $PaCO_2$ 40 mmHg: ratio 20, log 20 = 1.30103, pH 7.40.

Solutions & Colligatives

$$\Delta T_f = iK_fm \qquad \Delta T_b = iK_bm \qquad \Pi = iMRT$$

Raoult: $P_{soln} = X_{solvent}P°_{solvent}$. Water: K_f = 1.86 °C·kg/mol, K_b = 0.512 °C·kg/mol. *E.g.* 0.5 mol/kg NaCl: $\Delta T_f = 2 \times 1.86 \times 0.5 = 1.86$ °C.

Electrochemistry

$$E°_{cell} = E°_{cathode} - E°_{anode} \qquad \Delta G° = -nFE°$$

$$E = E° - \frac{RT}{nF}\ln Q = E° - \frac{0.0592}{n}\log Q$$

(25 °C, V). *E.g.* $E°$ = 1.1 V, n = 2, Q = 10: $E \approx$ 1.07 V.

Oxidation at the anode. **Galvanic:** $\Delta G < 0$, $E_{cell} > 0$, anode (−). **Electrolytic:** all three reversed.

CLINICAL NOTE Membrane potential (37 °C)

$$E_{ion} = \frac{61.5}{z}\log\frac{[X]_{out}}{[X]_{in}}\ \text{(mV)}$$

K^+ 4 mM out, 140 mM in: $E = 61.5 \times (-1.544068) \approx -95$ mV.

Organic Essentials

R/S: rank by atomic number at the first difference (CIP), point the lowest priority away, trace 1 → 2 → 3: clockwise *R*, counterclockwise *S* (reverse if it points toward you).

	S_N1	S_N2
Substrate	3° > 2°	methyl > 1° > 2°
Rate law	k[RX]	k[RX][Nu^-]
Nucleophile	weak is fine	strong
Solvent	polar protic	polar aprotic
Stereo	racemization	inversion

Amino acid pI (neutral side chain): pI = $\left(pK_{a1} + pK_{a2}\right)/2$; net charge + below pI, − above. *E.g.* 2.34 and 9.6: pI = 5.97.

Spectroscopy

Beer–Lambert

$$A = -\log T = \varepsilon lc$$

E.g. A = 0.5, ε = 5,000 L/(mol·cm), l = 1 cm: c = 100 µM.

CLINICAL NOTE Pulse oximetry

HbO_2 and Hb absorb 660 nm and 940 nm light differently; the absorbance ratio gives SpO_2.

IR band	cm^{-1}
O–H (alcohol), broad	3,400–3,650
O–H (carboxylic acid), broad	2,500–3,100
N–H, 1 or 2 peaks	3,300–3,500
C≡N / C≡C	2,100–2,260
C=O, strong & sharp	1,670–1,780

^{1}H NMR environment	δ (ppm)
alkyl C–H	0.7–1.8

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^{1}H NMR environment	δ (ppm)
CH_3–C=O, methyl ketone	2.0–2.4
H–C–O, alcohol, ether	3.3–4.5
vinylic =C–H	4.5–6.5
aromatic Ar–H	6.5–8.0
aldehyde CHO	9.7–10.0
carboxylic acid COOH, broad	11.0–12.0

^{1}H NMR: n neighboring H give $n + 1$ peaks; the integral is proportional to the number of H.

High-Yield Med Chem

Anion gap = $Na^+ - (Cl^- + HCO_3^-)$
Osm = $2\ Na^+ + Glu/18 + BUN/2.8$
$CaO_2 = 1.34 \times Hb \times SaO_2 + 0.003 \times PaO_2$
$t_{1/2} = 0.693 \times V_d / CL$

Glu, BUN in mg/dL; Hb in g/dL. Reference: AG 4–12 mEq/L, osm 275–295 mOsm/kg; intervals vary by lab and method.

EXAMPLE

- Na 140, Cl 100, HCO_3^- 14: AG = 26 mEq/L (high).
- Na 140, glucose 90, BUN 14: Osm = 290 mOsm/kg.
- Hb 15, SaO_2 0.98, PaO_2 100: $CaO_2 \approx$ 20.0 mL/dL, only 0.3 dissolved.
- V_d 50 L, CL 5 L/h: $t_{1/2} \approx$ 6.9 h.

CLINICAL NOTE Alveolar gas equation

$$P_AO_2 = F_IO_2\left(P_{atm} - P_{H_2O}\right) - P_aCO_2 / \text{RQ}$$

Sea level 760 mmHg; P_{H_2O} 47 mmHg at 37 °C; RQ ≈ 0.8.

Constants & Traps

$N_A = 6.022 \times 10^{23}$ mol^{-1} · $R = 8.314$ J/(mol·K) = 0.08206 L·atm/(mol·K) · $F = 96{,}485$ C/mol · $h = 6.626 \times 10^{-34}$ J·s · $c = 3.00 \times 10^8$ m/s · K_w (25 °C) $= 1.0 \times 10^{-14}$ · 1 atm = 760 mmHg = 101.325 kPa · ideal gas at 0 °C, 1 atm: 22.4 L/mol · 0 °C = 273.15 K · water $c = 4.184$ J/(g·°C)

TRAP

- Use kelvin; match J and kJ in ΔH, ΔS.
- M per L of solution; m per kg of solvent.
- Colligative effects count particles (i).
- Catalysts lower E_a; only T changes K.
- $\Delta G° = -RT \ln K$: with log, multiply by 2.303.
- At equilibrium $\Delta G = 0$ and $E_{cell} = 0$.

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